December 1, 2025

Advanced Statistics Project

1. Problem

Understanding which road and environmental conditions most contribute to accident occurrence is essential for designing effective prevention strategies and improving road-safety planning. Identifying the factors that increase accident risk allows institutions and operators to intervene more efficiently, prioritising resources where they can have the greatest impact.

In this project, we analyse accident risk using a dataset derived from the *Simulated Road Accident* framework, which reproduces realistic interactions between road geometry, weather, lighting, and contextual variables. Our objectives are twofold: first, to study accident risk as a continuous outcome in order to quantify how different features influence it; second, to classify road conditions into high-risk and low-risk categories to provide a more operational view of dangerous scenarios.

2. Data

The analysis is based on a synthetic dataset generated from the *Simulated Road Accident* framework. The dataset contains road characteristics (e.g., type, curvature, speed limits), environmental conditions (lighting, weather, time of day), and contextual variables such as holidays and school seasons. Its size and completeness make it suitable for both regression and classification tasks. Although synthetic, the dataset reflects realistic behavioural patterns and has been used in predictive modelling contexts, including Kaggle-based applications. In our project, we use the complete dataset provided, which includes the target variable *accident risk*. Since the data contain no missing values and are already well structured, only minimal preprocessing is required before applying the train–test split.

The dataset is relevant for our problem because it captures diverse conditions under which accidents become more likely, enabling a meaningful evaluation of statistical learning methods.

3. Method

We approach the problem using both regression and classification frameworks. For the regression task, we model the continuous accident risk using Ordinary Least Squares (OLS) as a baseline, together with penalised linear models, Ridge and Lasso, to address multicollinearity, stabilise coefficient estimates, and reduce overfitting in the presence of numerous one-hot encoded features. Since the response variable is bounded in $[0, 1]$, we also estimate a Beta Regression model, which provides a more appropriate likelihood-based framework for rates and proportions while allowing a direct comparison with linear methods.

To complement the regression analysis, we reformulate the problem as a binary classification task by thresholding accident risk at its median[1] , producing balanced "high-risk" and "low-risk" classes. This classification perspective allows us to evaluate how well different models distinguish dangerous road conditions. Logistic Regression with L2 regularisation serves as a

[1]The median ensures a 50/50 class split, which stabilises training and avoids class imbalance issues that may arise if using the mean.

transparent linear benchmark, while a Random Forest classifier is used to capture non-linear effects and interactions among road, weather, and contextual variables. Contrasting these two approaches highlights the trade-off between interpretability and predictive flexibility.

Overall, combining linear, penalised, and distribution-aware regression methods with both linear and non-linear classifiers enables a comprehensive assessment of accident risk under different modelling assumptions and helps identify the variables that most consistently influence it.

4. Implementation

All analyses were conducted in Python using `pandas`, `scikit-learn`, `seaborn`, and `shap`. Categorical variables were transformed through one-hot encoding to ensure compatibility with linear models, while numerical features were standardised using `StandardScaler`, a necessary step for Ridge and Lasso since their penalties depend on feature magnitude. The complete dataset was then split into training and test sets (80–20), using stratification in the classification task to preserve the proportion of high- and low-risk observations.

For the regression task, OLS was fitted as an initial baseline, and diagnostic tools such as the Variance Inflation Factor (VIF) were computed to quantify multicollinearity. Ridge and Lasso models were then implemented using 5-fold cross-validation to select the optimal regularisation parameter λ, and coefficients were extracted to compare shrinkage patterns and estimator stability. In addition, a Beta Regression model was estimated using a logit link, providing a likelihood-based approach suitable for outcomes bounded in $[0, 1]$ and enabling direct comparison with linear models.

For the classification task, the accident risk variable was thresholded at its median to create balanced high-risk and low-risk classes. Logistic Regression with L2 regularisation served as the linear benchmark, while a Random Forest classifier was tuned using `RandomizedSearchCV` over parameters such as number of trees, maximum depth, and minimum samples per split. Performance metrics were computed on the test set, and confusion matrices were examined to analyse misclassifications. To enhance interpretability, we computed permutation importance, SHAP values, and partial dependence plots, allowing a detailed assessment of variable influence and non-linear effects.

The resulting pipeline ensures a consistent and transparent framework for evaluating linear, penalised, and distribution-aware regression methods alongside both linear and non-linear classifiers, enabling a rigorous assessment of accident risk modelling.

5. Evaluation

The performance of the regression models was assessed using standard metrics for continuous outcomes. Mean Squared Error (MSE) served as the primary criterion, as it penalises larger deviations and is well suited for evaluating probabilistic predictions. Mean Absolute Error (MAE) was also computed to provide a more robust assessment in the presence of extreme values, while the coefficient of determination (R^2) quantified the proportion of variability explained by each model. Test-set metrics were compared against training results to detect potential overfitting and evaluate the effect of regularisation.

For the classification task, evaluation focused on metrics appropriate for balanced classes, since the median threshold yields two groups of similar size. Accuracy and F1-score were used as primary indicators of predictive performance, with the latter emphasising the model's ability to correctly identify high-risk cases. Balanced accuracy was included to account for potential asymmetries between classes, and ROC-AUC provided a threshold-independent measure of

discriminative ability. Confusion matrices were examined to highlight systematic misclassifications, particularly false negatives, which are especially important in a risk-detection setting.

Taken together, these evaluation criteria allow for a consistent comparison across regression and classification methods, offering a clear basis for assessing predictive performance and understanding model behaviour.

6. Interpretation

The combined analysis across EDA, regression, and classification provides a coherent view of the mechanisms shaping accident risk. Exploratory findings indicated that adverse weather, poor lighting, and road curvature consistently correlate with higher risk, suggesting that accident probability rises when structural and environmental difficulties accumulate. The modelling stage confirms and refines these insights.

Across all regression models, a consistent pattern emerges: road curvature, lighting conditions, and weather are the main determinants of accident risk. Curvature shows the strongest effect, as more demanding road geometry increases driver workload and reduces reaction time. Lighting and weather amplify this effect: nighttime, fog, and rain substantially raise the likelihood of an accident even when road structure remains unchanged. The fact that OLS, Ridge, Lasso, and Beta Regression achieve almost identical predictive performance ($R^2 \approx 0.80$ for all models) indicates that the underlying relationships are stable and largely linear. Regularisation mainly improves the stability of coefficient estimates, while Lasso highlights that only a limited subset of features truly drives accident risk. Beta Regression reinforces these conclusions by modelling the bounded nature of the response without altering the core message: accident risk is shaped by the combined influence of geometry and visibility.

The classification results complement this view. Logistic Regression captures the main trend but misses important interactions, which explains its moderate accuracy. The Random Forest, by contrast, captures the complex ways in which curvature, lighting, and weather reinforce each other, achieving much stronger predictive power (ROC-AUC ≈ 0.96) and reducing false negatives, an essential property when identifying potentially dangerous situations. The variables identified as most important by the Random Forest mirror those highlighted in the regression analysis, showing that the conclusions are robust across methodological frameworks.

SHAP values further clarify these mechanisms. They reveal that curvature has the largest impact on predicted risk, followed by lighting and weather conditions. The plot below illustrates how risk increases sharply when challenging road geometry coincides with poor visibility, confirming that accidents arise from the interaction of multiple stressors rather than from isolated factors.

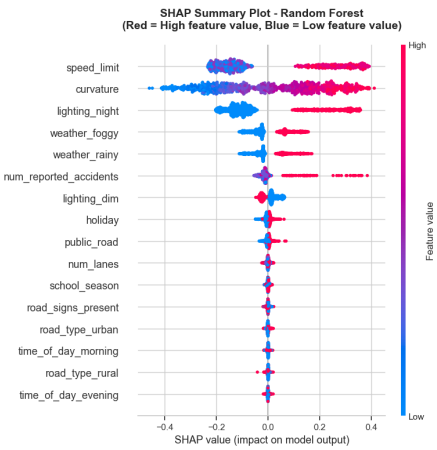

Figure 1: SHAP feature importance (Random Forest): curvature, lighting, and weather emerge as the dominant predictors.

Taken together, these results indicate that accident risk follows a clear and interpretable structure. Road geometry and visibility conditions are the primary drivers of danger, while weather further amplifies their effects. Linear models quantify these relationships, penalised models stabilise them, and non-linear classifiers capture the interactions that shape high-risk scenarios. These insights point to concrete mitigation strategies, such as improving lighting on curved segments, adjusting speed limits during poor weather, and deploying targeted warning systems, that directly address the conditions identified as the main contributors to accident risk.

Annex(*)

In this section we report additional visualisations that support the main analysis but are not strictly required for the interpretation of the results.

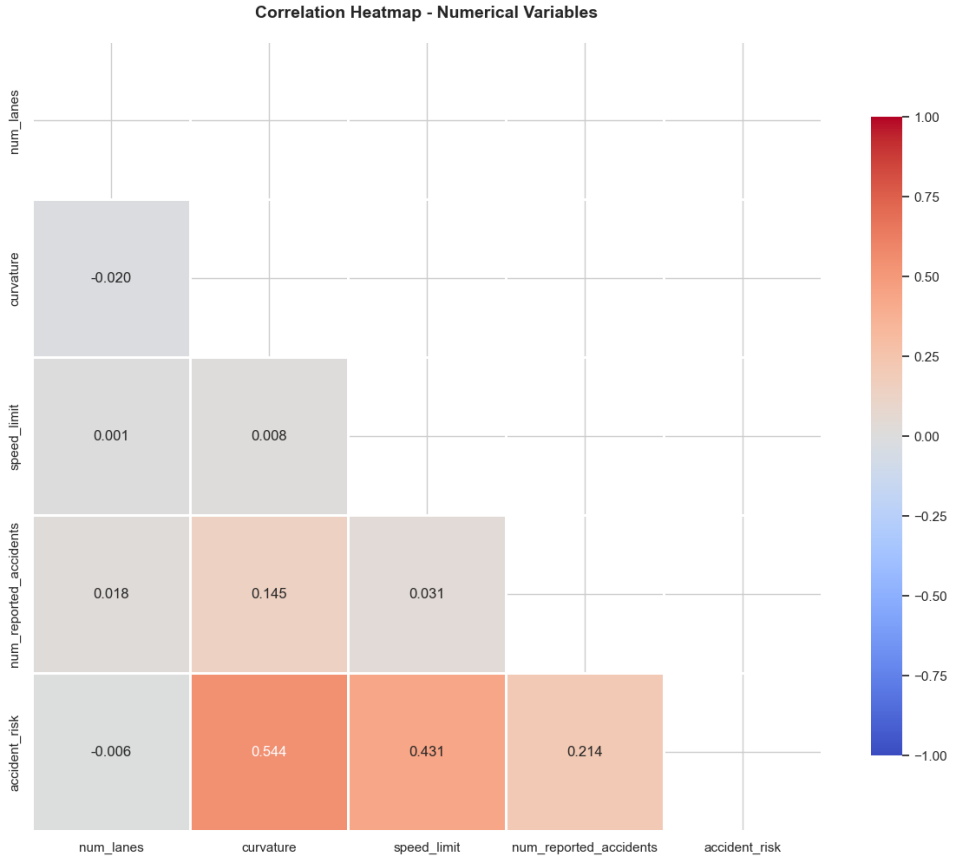

Figure 2: Correlation heatmap of numerical variables. The plot highlights the presence of moderate to strong correlations among structural features such as curvature, speed limit, and number of lanes, suggesting potential multicollinearity in the regression setting.

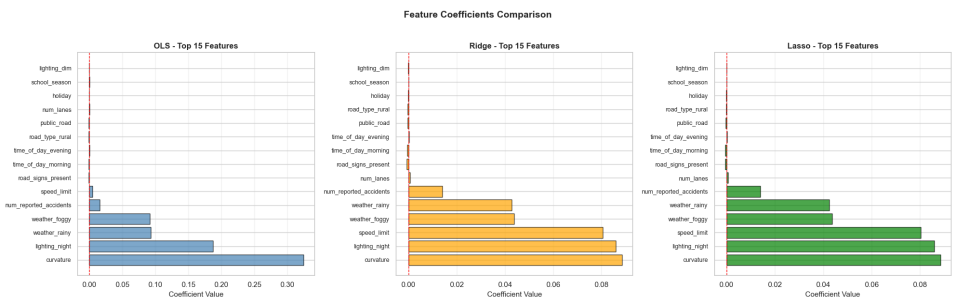

Figure 3: Comparison of the top coefficients estimated by OLS, Ridge, and Lasso. The plot illustrates how Ridge stabilises the magnitude of correlated predictors, while Lasso performs feature selection by shrinking several coefficients to zero.

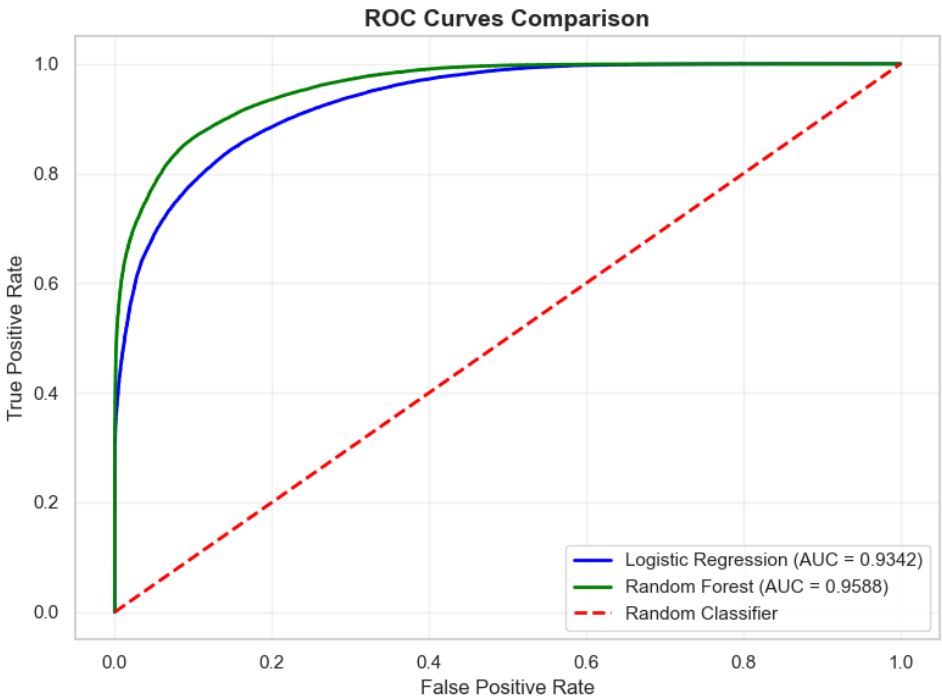

Figure 4: ROC curve comparison between Logistic Regression and Random Forest. The Random Forest achieves a higher AUC, indicating superior ability to distinguish between high-risk and low-risk conditions across different thresholds.